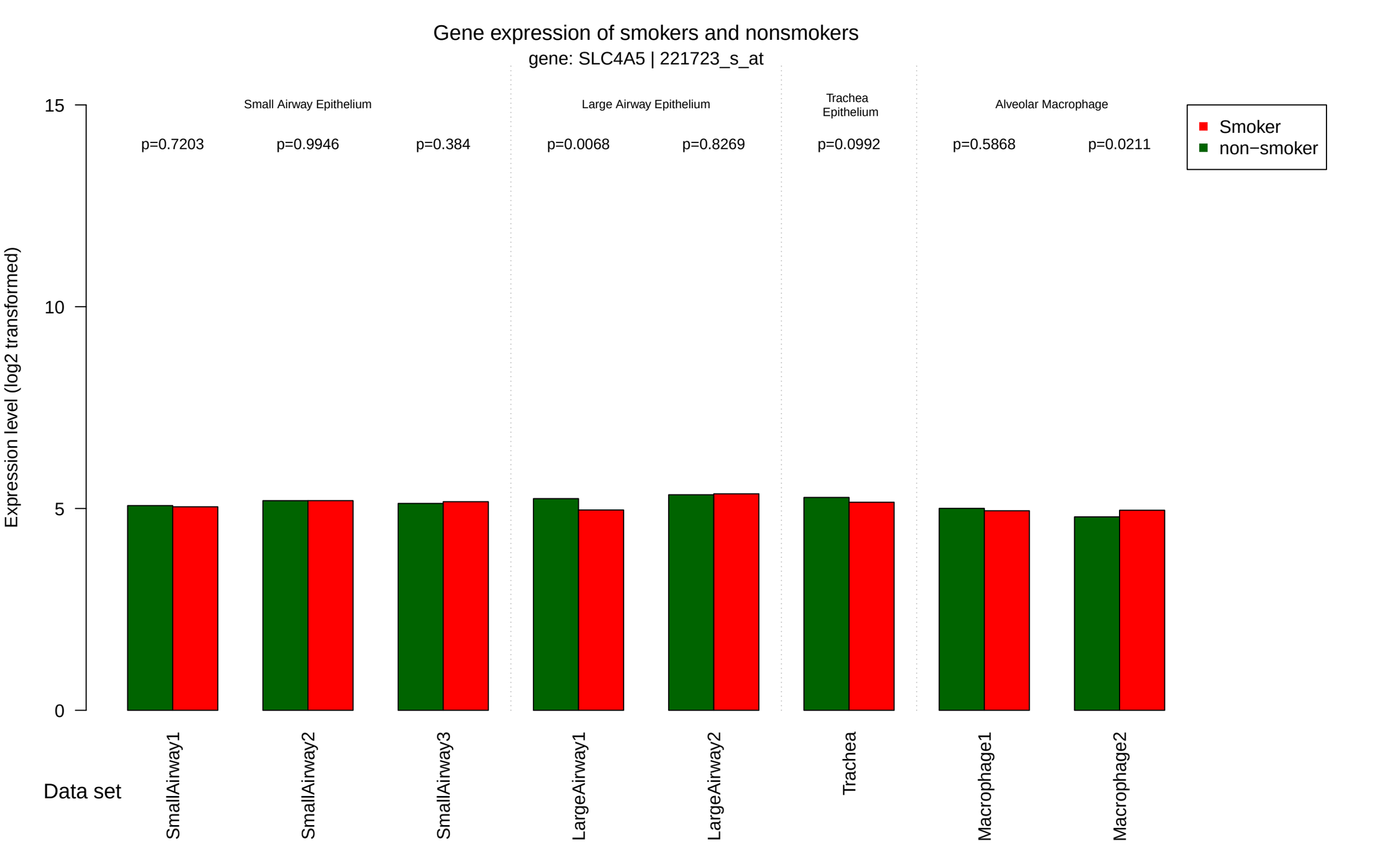
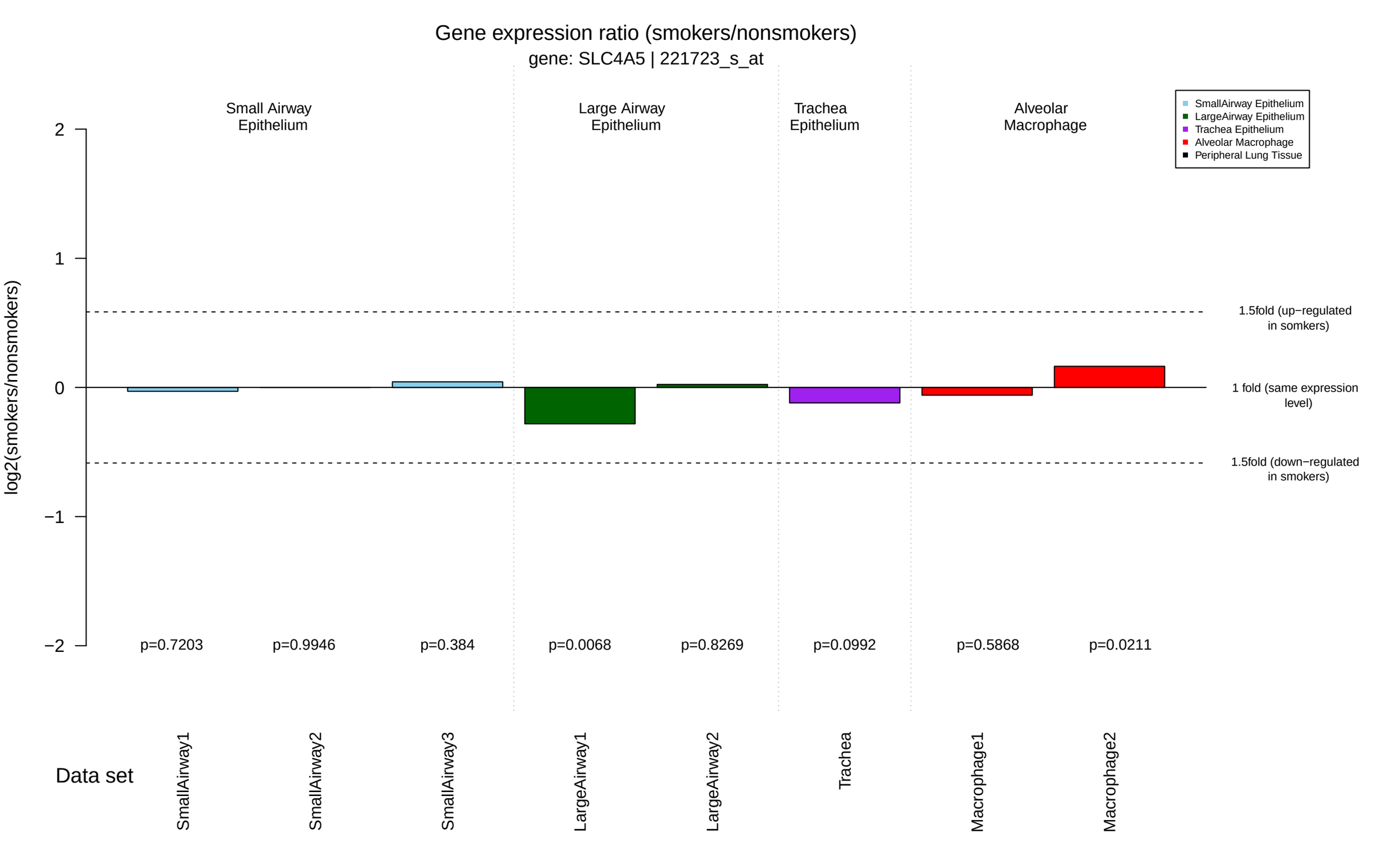
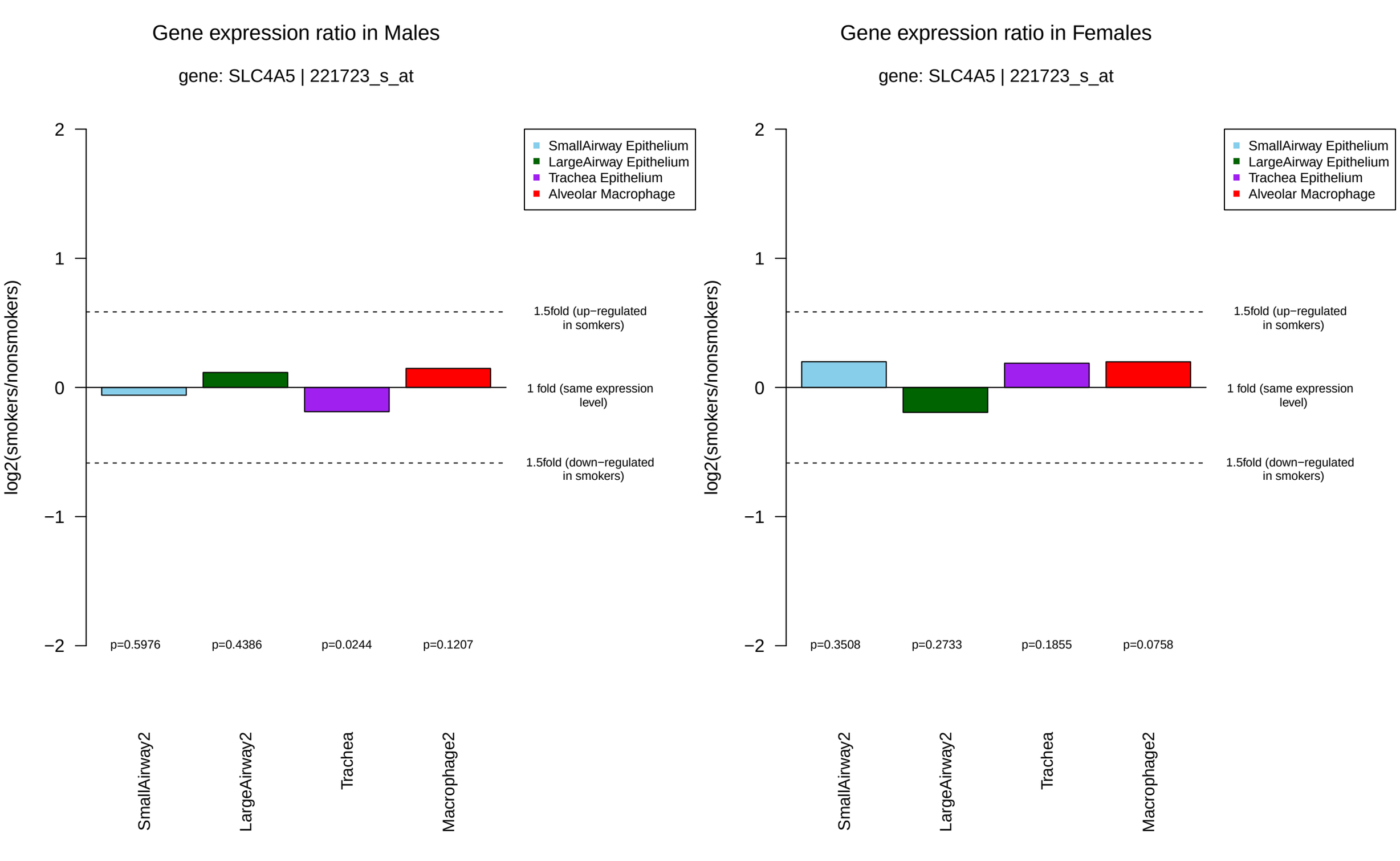
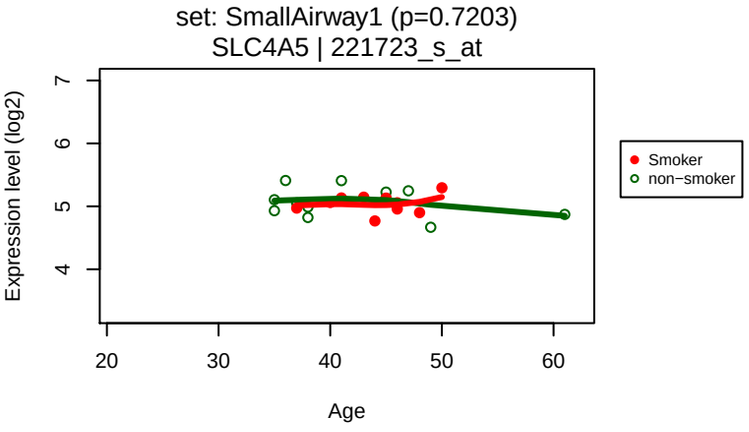
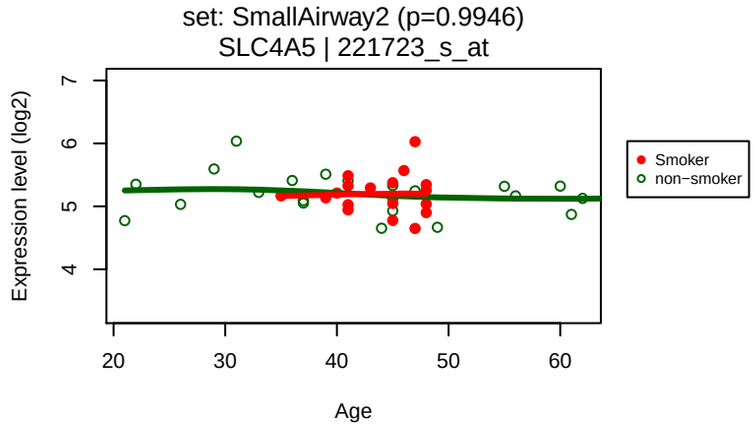
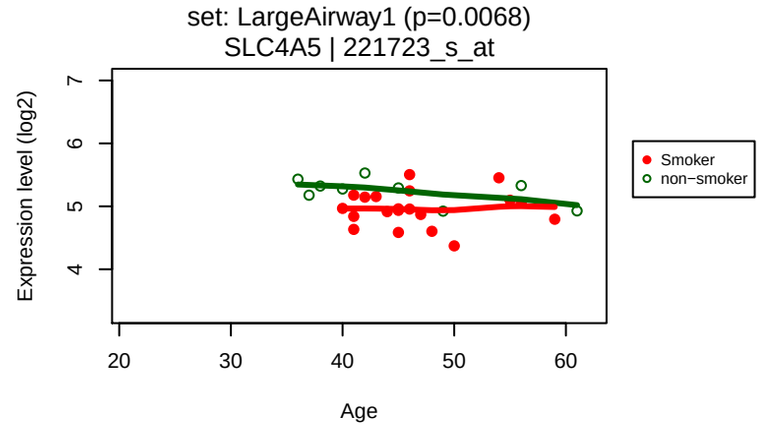

Gene expression of smokers and nonsmokers
gene: SLC4A5 | 221723_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7203
p=0.9946
p=0.384
p=0.0068
p=0.8269
p=0.0992
p=0.5868
p=0.0211
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

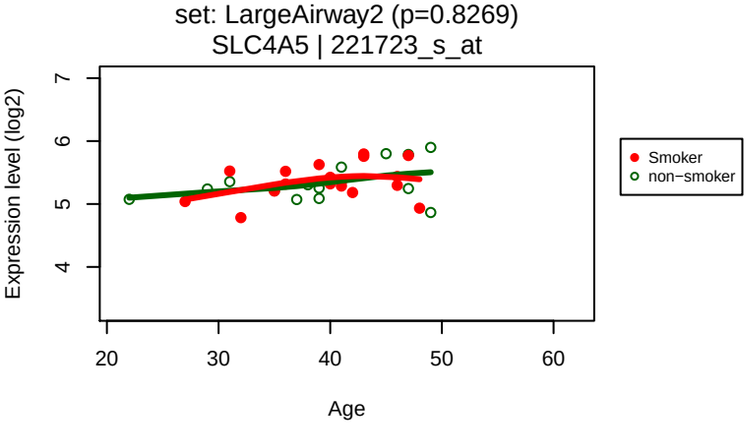

Gene expression ratio (smokers/nonsmokers)
gene: SLC4A5 | 221723_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7203
p=0.9946
p=0.384
p=0.0068
p=0.8269
p=0.0992
p=0.5868
p=0.0211
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

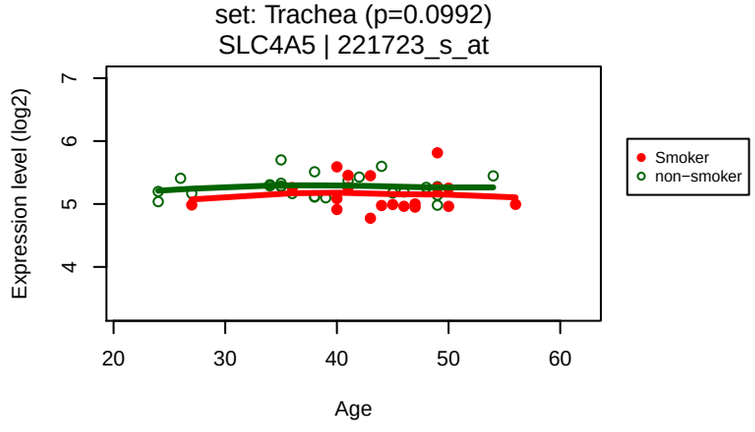

Gene expression ratio in Males
gene: SLC4A5 | 221723_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5976
p=0.4386
p=0.0244
p=0.1207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC4A5 | 221723_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3508
p=0.2733
p=0.1855
p=0.0758
SmallAirway2
LargeAirway2
Trachea
Macrophage2

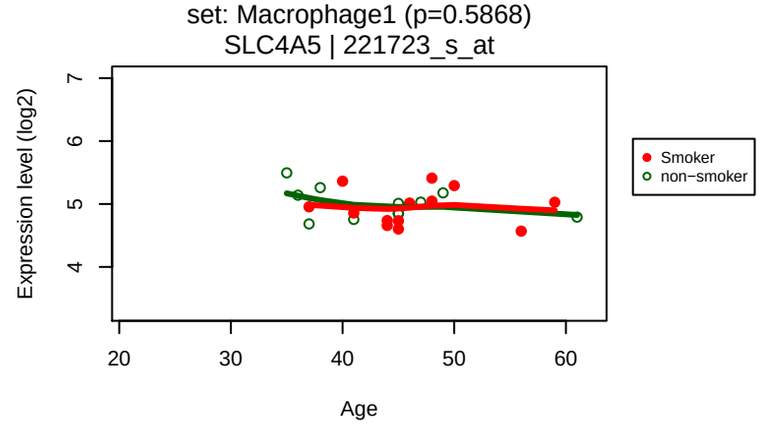

set: SmallAirway1 (p=0.7203)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker

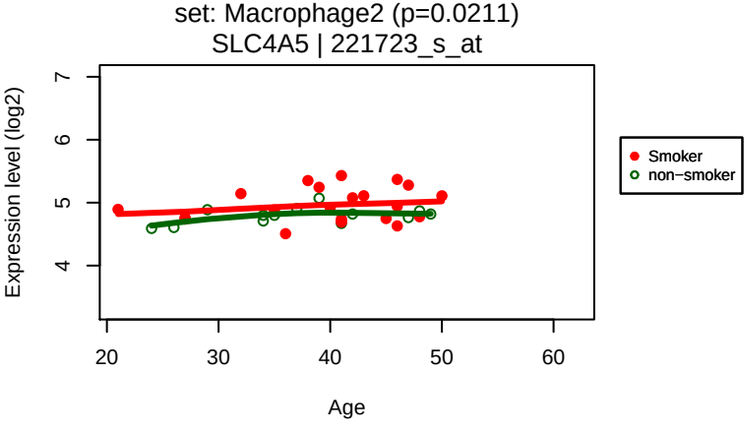

set: SmallAirway2 (p=0.9946)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker

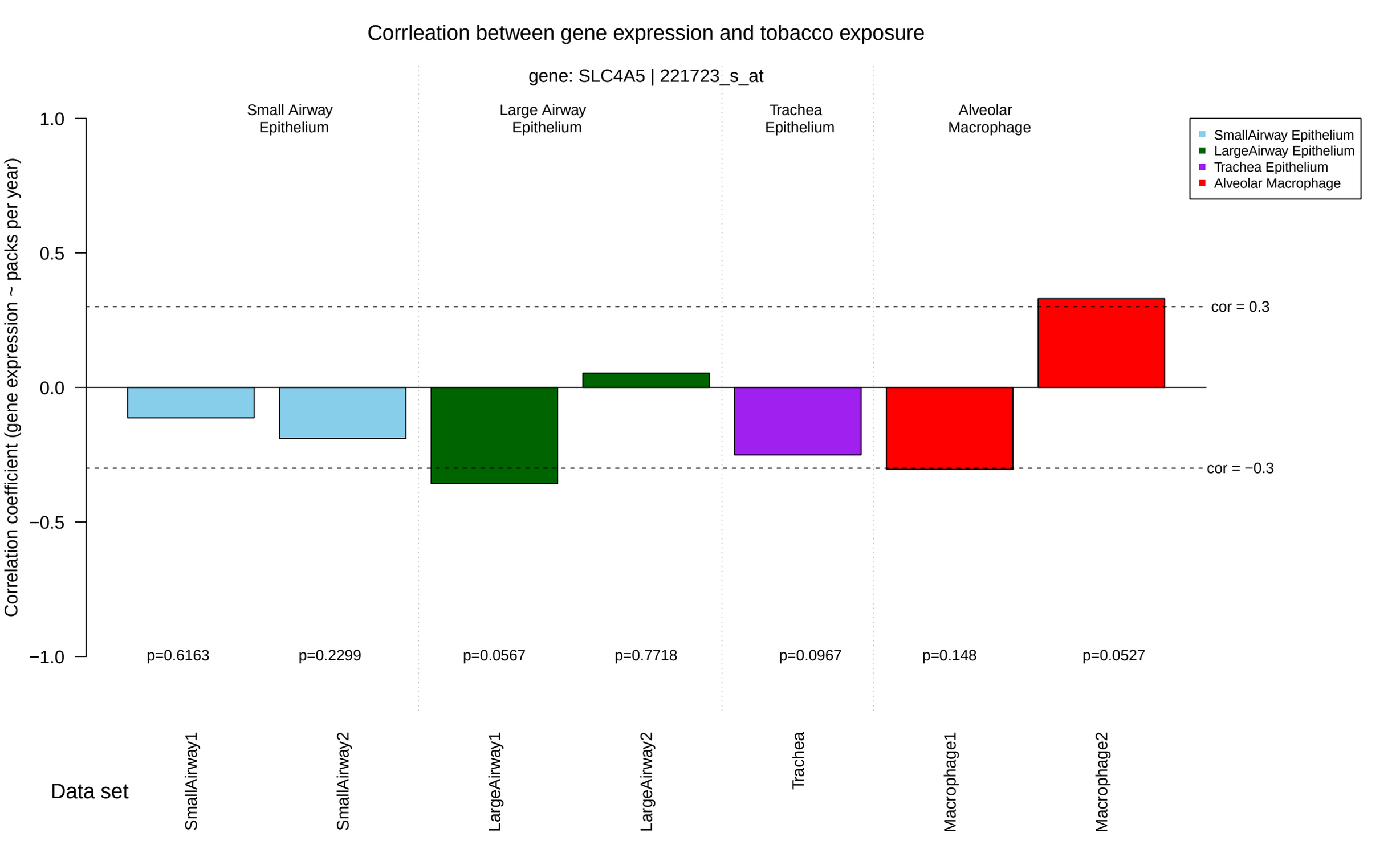

set: LargeAirway1 (p=0.0068)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8269)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0992)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5868)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0211)
SLC4A5 | 221723_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC4A5 | 221723_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6163
p=0.2299
p=0.0567
p=0.7718
p=0.0967
p=0.148
p=0.0527
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2